LESSON PLAN

COMPUTER (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Radar system introduction to radar system, primary and secondary radar system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Radar system introduction to radar system, primary and secondary radar system
5. Specific Objectives	26.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Radar Block Diagram and Signal Flow; Safety: No live equipment handling; Supervision: Instructor monitors PPT navigation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the full form of RADAR?
(b) Define the basic principle of communication theory.
(c) List two prerequisites for this lesson.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic Theory of Communication • Radar System Introduction • Radar Block Diagram Components • Pulse Transmission and Reception • Signal Processing Basics	L & Ds	PPT, IB	15 min
(b)	• Primary Radar System Principles • Secondary Radar System Principles • Transponder Functionality • Interrogation and Reply Cycle • Comparison of Primary and Secondary	L & Ds	PPT, PDF	20 min
(c)	• Radar Range Equation Factors • Antenna Beamwidth and Gain • Clutter and Noise Sources • Display Types: PPI and A-Scope • Summary of System Architecture	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the fundamental difference between primary and secondary radar systems.
(b) Describe the role of the transponder in a secondary radar system.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of this lesson bridge to the study of EMI and EMC in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Basics of EMI and EMC. Types of testing.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min